

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester II Examination, 2013-14

Zoology**Paper: ZOB-201 — Animal Form and Function and Elementary Biochemistry**

Time: Three hours

Full Marks: 70

Note: Answer three questions from Section-A and three questions from Section-B. Question No. 1 in both sections is compulsory. Use a separate answer book for each section. The figures in the right-hand margin indicate marks.

SECTION – A**(Animal Form and Function) (Marks: 35)****Question 1.** Answer the following parts:

- (a) Define the following terms: [3 × 1 = 3]
- (i) Deposit feeding
 - (ii) Longitudinal binary fission
 - (iii) Dorsal ocelli.
- (b) State whether the following statements are True or False giving reasons: [4 × 2 = 8]
- (i) Ammonia is a major nitrogenous waste in aquatic animals.
 - (ii) Each hair cell in fish ear has single kinocilium.
 - (iii) Spiracle regulates tracheal respiration in aquatic insects.
 - (iv) Parthenogenesis occurs in mammals.

Question 2. Answer the following parts:

- (a) Describe the structure and function of vertebrate lungs. Explain the double ventilation process in bird. [6]
- (b) What is nerve net? Explain the pattern of nervous system in invertebrates with labeled diagrams. [6]

Question 3. Answer the following parts:

- (a) Draw a neat labeled diagram of an ommatidium. Describe the mechanism of photoreception in insect. [6]
- (b) Explain the structure and function of primary and accessory sex organs of mammals. [6]

Question 4. Write short notes on the following: [3 × 4 = 12]

- (a) Intracellular digestion
- (b) Parental care in amphibians
- (c) Sympathetic nervous system.

SECTION – B
(Elementary Biochemistry) (Marks: 35)

Question 1. Answer the following parts:

- (a) Select and write the most correct alternative for the following: [2 × $\frac{1}{2}$ = 1]
- (i) Which one of the following enzymes is a transferase?
- (1) Glucokinase
 - (2) Malate dehydrogenase
 - (3) Acetylcholine esterase
 - (4) Fumarase
- (ii) Which one of the following is the principal component of the bone?
- (1) Fibroin
 - (2) Keratin
 - (3) Collagen
 - (4) Elastin
- (b) Select if the following statements are True or False. Justify in 2–3 lines each: [3 × 2 = 6]
- (i) Ester linkages of triglycerides can be hydrolyzed by acid or alkali.
- (ii) There is an asymmetrical distribution of charge in water molecule. Hydrogen nucleus draws electrons from oxygen nucleus which results in a net positive charge.
- (iii) Peptidoglycan is a heteropolysaccharide of alternating NAG-NAM of β (1-4) linkages in the bacterial cell wall.
- (c) Define the following: [2 × 1 = 2]
- (i) Anomers
- (ii) Conformation.
- (d) Draw the structures of the following: [2 × 1 = 2]
- (i) 2'-deoxy thymidine
- (ii) Phenyl alanine.

Question 2. Answer the following parts:

- (a) Describe the process of transcription in prokaryotes with well-labeled diagrams. [6]
- (b) What is Michaelis Constant (K_m)? Illustrate the mechanism of enzyme action by explaining the binding of substrate to enzyme. [6]

Question 3. Answer the following parts:

- (a) Describe diagrammatically the process of replication in *E. coli*. [6]

(b) Differentiate between reducing and non-reducing sugars. [6]

Question 4. Write short notes on the following: [3 × 4 = 12]

- (a) Unsaturated fatty acid
- (b) Peptide bond
- (c) Applications of recombinant DNA techniques.

[Click Here](#)

[Click Here](#)

[Click Here](#)